QUANT

Venn Diagrams

Assignment 04

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Venn Diagram Online Assignment -04

1. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many like both coffee and tea?

1. 18 2. 19 3. 15 4. 12

2. A Survey shows that 76% of the Indians like oranges, whereas 62% like bananas. What percentage of the Indians like both oranges and bananas?

1. 15 2. 22 3. 25 4. Cannot be determined

3. In a group of 950 persons, 750 can speak Hindi and 460 can speak English. Find how many can speak both Hindi and English if each person speak at least one of the language.

1. 150 2. 175 3. 160 4. None of these

4. In a group of 50 persons, 14 drink tea but not coffee and 30 drink coffee but not tea and each person drink at least one of the two drinks. How many drink both coffee and tea?

1. 6 2. 8 3.9 4.11

5. In a survey of 60 people, it was found that 25 people read newspaper *H*, 26 read newspaper T 26 read newspaper *I*, 9 read both *H* and *I*, 11 read both H and T, 8 read both T and *I*, 3 read all three newspapers. Find the numbers of people who read at least one of the newspapers.

1. 44 2.32 3. 48 4. 52

6. Of the members of three athletic teams in a certain school, 21 are in the basketball team, 26 in hockey team and 29 in the football team. 14 play hockey and basket ball, 15 play hockey *and* football, 12 play football and basketball and 8 play all the three games. How many members are there in all?

1. 41 2.42 3. 43 4. 44

7. A survey of 500 television viewers produced the following information; 285 watch football, 195 watch hockey, 115 watch basketball, 45 watch football and basketball, 70 watch football and hockey, 50 watch hockey and basketball, 50 do not watch any of the three games. How many watch all the three games? How many watch exactly one of the three games?

1. 21 2. 20 3. 33 4. 38

8. In a survey of 100 persons it was found that 28 read magazine *A*, 30 read magazine *B*, 42 *read* magazine C, 8 read magazines *A* and B, 10 read magazines *A* and C, 5 read magazines B *and C* and 3 read all the three magazines. Find how many read none of three magazines?

1. 30 2.27 3. 23 4. 20

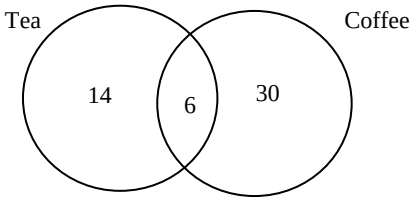
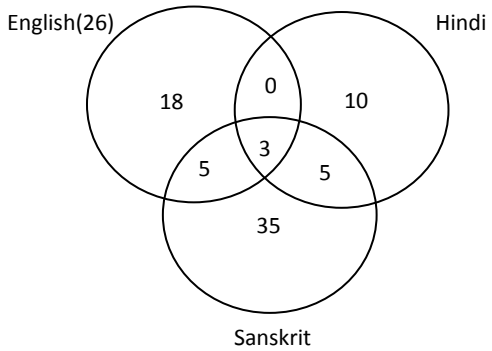
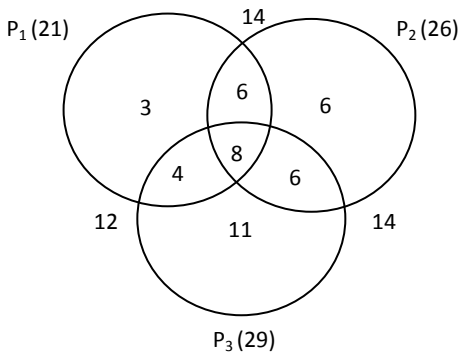
9. In a survey of 100 students, the number of students studying the various languages were found to be: English only, 18, English but not Hindi 23, English and Sanskrit 8, English 26, Sanskrit 48, Sanskrit and Hindi 8, no language 24. Find How many students were studying Hindi?

1. 18 2. 15 3. 13 4. 14

10. In a survey it was found that 21 persons liked product P_1, 26 liked product P_2 and 29 liked product P_3. If 14 persons liked products P_1 and P_2 ; 12 persons liked product P_3 and P_1 ; 14 persons liked products P_2 and P_3 and 8 liked all the three products. Find how many liked product P_3 only.

1.10 2.8 3. 11 4.14

Answer Key & Explanations

Sr. No.	Key	Explanation
1	2	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ $70 = 37 + 52 - n(A \cap B)$ $n(A \cup B) = 37 + 52 - 70$ $= 89 - 70 = 19$
2	4	Cannot be determined as nowhere it is given that each Indian like at least one of the fruits
3	3	$950 = 750 + 460 - n(A \cap B)$ $950 = 1210 - n(A \cap B)$ $n(A \cap B) = 1210 - 950 = 160$
4	1	 As shown above 6 drink both tea and coffee
5	4	$n(A \cup B \cup C) = n(A) + nB + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$ $= 25 + 26 + 26 - 9 - 11 - 8 + 3$ $= 77 + 3 - 28 = 52$
6	3	$n(A \cup B \cup C) = 21 + 26 + 29 - 14 - 15 - 12 + 8$ $= 84 - 41 = 43$
7	2	The person's who water at least one of the game = 500 - 50 = 450 $450 = 285 + 195 + 115 - 45 - 70 - 50 + n(A \cap B \cap C)$ $450 = 595 - 165 + n(A \cap B \cap C)$ $450 = 430 + n(A \cap B \cap C)$ $n(A \cap B \cap C) = 450 - 430 = 20$
8	4	$n(A \cap B \cap C) = 28 + 30 + 42 - 8 - 10 - 5 + 3$ $= 100 - 23 + 3 = 80$ ∴ The person who read none = 100 - 80 = 20
9	1	Total no of students who were study at least m language = 100 - 24 = 76  As shown above students speaking Hindi = 10 + 5 + 3 = 18
10	3	 As shown above P_3 only are 11.